

Faustina Asante

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PROFILE

Mathematical statistician and biostatistics researcher with 5+ years of experience applying advanced statistical modeling, Bayesian methods, survival analysis, and data science to complex health and population data, with strong programming expertise in R, SAS, Python, and SQL.

EDUCATION

Ph.D. Mathematical Sciences - Statistics | Northern Illinois University, DeKalb, IL

Expected Dec 2027

M.S. Artificial Intelligence in Business | Northern Illinois University, DeKalb, IL

Expected Dec 2026

M.S. Statistics | Northern Illinois University, DeKalb, IL

May 2024

B.Sc. Mathematics and Statistics | University of Cape Coast, Ghana

May 2019

RESEARCH AND PROFESSIONAL EXPERIENCE

Research Analyst | NIU Foundation– Advancement Operations, DeKalb, IL

June 2024 – Present

- Developed and maintained 10+ Power BI dashboards, scorecards, and automated reporting pipelines; reduced manual reporting time by approximately 40% and expanded leadership access to operational KPIs.
- Built Gap-to-Goal and Chicagoland regional dashboards to track fundraising progress and geographic engagement, informing advancement teams' planning and outreach using Power BI and ArcGIS.
- Utilized CRM software such as Salesforce, Millennium, Ever True, Alumni Finder, LexisNexis, Thank View, and Magnet for data accuracy and to ensure data integrity.
- Conducted exploratory data analysis (EDA) and trend analysis to identify patterns, measure performance, and provide recommendations for process improvement and operational efficiency.

Graduate Assistant and Statistical Consultant | NIU Department of Statistics, DeKalb, IL **Aug 2022 – May 2024**

- Manage analytical research for 15+ academic, health, nonprofit, and business clients, translating scientific questions into testable hypotheses, statistical designs, analysis plans, and interpretable deliverables.
- Apply logistic and linear regression, ANOVA, hypothesis testing, survival analysis, Bayesian methods, causal inference, simulation, and model diagnostics in R and SAS to address applied research questions.
- Analyze administrative claims, electronic medical records, surveillance data, and other health-related sources to support epidemiologic, clinical, and outcomes research.
- Conduct literature reviews and prepare technical reports, scientific manuscripts, presentations, and dashboards; explain assumptions, uncertainty, limitations, and findings to investigators with varied quantitative backgrounds.
- Standardize SQL validation and Python data-cleaning workflows, document analytical processes, and maintain reproducible methods that strengthen transparency and scientific rigor.

Data Analyst | Ghana Statistical Service, Accra, Ghana

Oct 2020 – Jul 2022

- Upskilled and mentored over 200 field statisticians and data officers in data cleaning, automated workflows, data visualization, and large-dataset manipulation using CPro, SPSS, R, and Advanced Excel.
- Automated recurring preparation workflows in Stata and Excel, reducing manual processing time by approximately 50% and accelerating statistical releases.
- Engineered robust ETL (Extract, Transform, Load) pipelines and data integration workflows to consolidate administrative records, and inter-agency research sources into centralized analytical databases.
- Conducted comparative risk assessments and epidemiological data analysis across diverse industries to identify workplace hazards, evaluate compliance operations, and establish evidence-based mitigation strategies for national labor studies.

PUBLICATIONS

- Adjei, A. K., Kataboh, P. K., **Asante, F.**, & Boyetey, E. (2022). A binary logistic regression analysis on the factors associated with high blood pressure and its related heart issues. *Science Journal of Applied Mathematics and Statistics*, 10(3), 38–44.
- Adjei, A. K., Kataboh, P. K., **Asante, F.**, & Gebechukwu, J. A. (2022). A dummy variable regression analysis of the impact of social media on university students' academic performance. *International Journal of Scientific and Research Publications*, 12(5), 53–57.

RESEARCH AND TECHNICAL EXPERTISE

Statistical Methods	Nonparametric regression • Bayesian inference • survival analysis • longitudinal and time-series methods • causal inference • multivariable regression • ANOVA • simulation • model selection and diagnostics
Biomedical Research	Cardiovascular and population-health analytics • epidemiologic studies • observational data
Programming and Data	R • SAS • Python • SQL • Stata • SPSS • ETL • large-scale data validation • reproducible workflows • data visualization • Power BI • Excel/VBA
Scientific Collaboration	Study design • analytical planning • literature synthesis • technical writing • manuscripts • research presentations • stakeholder consultation • interdisciplinary teamwork

SELECTED BIOMEDICAL AND PUBLIC HEALTH RESEARCH

- **Determinants of Heart Attack Among U.S. Adults, BRFSS 2022** - Analyzed national health-surveillance data; developed and compared multivariable logistic and probit models, evaluated demographic, behavioral, disability, and social predictors, and communicated implications for cardiovascular prevention.
- **Healthcare Worker Burnout Intervention Evaluation** - Evaluated engagement, well-being, and burnout indicators using quantitative methods and translated findings into recommendations for workforce wellness.
- **Smoking and Lung Cancer Outcomes Study** - Examined observational health data to identify smoking-related risk factors and support evidence-based prevention recommendations.
- **Obesity and Heart Disease Risk Factors Study** - Applied epidemiologic and multivariable methods to characterize behavioral and demographic cardiovascular-risk patterns and identify higher-risk populations.

Doctoral DISSERTATION RESEARCH

A Nonparametric Circular Linear Model for Time Series Data

- Develop a Bayesian nonparametric framework for data involving periodic or directional structure and a linear response, using natural cubic splines for linear trend and circular cubic splines for periodic effects.
- Design simulation studies to evaluate estimation and predictive performance and compare the proposed approach with parametric circular–linear models and classical time-series methods.
- Apply the framework to real-world atmospheric CO₂ and monthly airline-passenger data to estimate nonlinear trend and seasonality while preserving circular structure.
- Extend the methodology toward survival outcomes with circular covariates, motivated by biomedical processes whose risk may vary by time of day, season, or other recurring cycles.

LEADERSHIP AND HONORS

Graduate Senator, NIU Student Government Association	Aug 2025–May 2026
International Association of Black Actuaries Scholarship	Aug 2026
Outstanding Women and Gender Advocacy Student Award	April 2026
President’s Student Service and Leadership Award	April 2026
Dr. Carla Montgomery Memorial Award for Women in STEM	April 2025
Carol J. Feltz Memorial Scholarship	May 2023

PROFESSIONAL AFFILIATION

Member, American Statistical Association
Member, American Mathematical Association
Member, Association of Women in STEM
Member, American Association of University Women
Member, International Association of Black Actuaries
Member, Women in MATH
Member, P.E.O